

SNF-Bench: Separating Static Drift from Natural Flow in Long-Horizon Fixed-Camera Video Generation

Supplementary Material

This supplement records the material supporting the main paper without being required to follow its argument: the region-partition procedure and its sensitivity study, complete audit tables for both tracks at every horizon, measurement coverage, protocols for the two complementary perceptual checks, and additional qualitative material. The text-conditioned and image-conditioned tracks are reported here at equal depth. Section numbering is independent of the main paper.

S1. Formal Factor Definitions

The definitions below are those referenced from the main paper’s protocol section. Notation follows it: I_t is frame t of a T -frame sequence, Ω_{static} and Ω_{flow} are the region masks, $\mathbf{u}_t$ is the optical flow from I_t to I_{t+1} , and $\mathcal{E}$ and $\mathcal{L}$ are the early and late windows.

Feature-aligned Background Drift. For matched ORB pairs $\mathcal{M}_t$ between the first frame and frame t ,

$$D_{\text{feat}}(t) = \text{median}_{(p,q) \in \mathcal{M}_t} \|p - q\|_2, \\ \text{fBD} = \frac{100}{d} \cdot \frac{1}{|\mathcal{L}|} \sum_{t \in \mathcal{L}} D_{\text{feat}}(t) \quad (\downarrow), \quad (\text{S1})$$

with d the frame diagonal, so fBD is a percentage of image size.

Normalized Background Flow.

$$\text{NBF} = \frac{10^3}{W \Delta t} \cdot \frac{1}{T-1} \sum_{t=1}^{T-1} \frac{1}{|\Omega_{\text{static}}|} \sum_{\mathbf{x} \in \Omega_{\text{static}}} \|\mathbf{u}_t(\mathbf{x})\|_2 \quad (\downarrow), \quad (\text{S2})$$

with W the frame width and Δt the interval of the sampled pairs actually used for flow estimation, so the quantity is independent of both the native frame rate and the sampling policy.

Global-motion compensation. With $T_t(\mathbf{x}) = s_t R_t \mathbf{x} + \mathbf{b}_t$ the robust similarity transform fitted to static-region correspondences and $\mathbf{d}_t(\mathbf{x}) = T_t(\mathbf{x}) - \mathbf{x}$ the displacement field it induces,

$$\tilde{\mathbf{u}}_t(\mathbf{x}) = \mathbf{u}_t(\mathbf{x}) - \mathbf{d}_t(\mathbf{x}). \quad (\text{S3})$$

Motion-Compensated Foreground Flow and Flow Persistence.

$$\text{MCFF}(\mathcal{W}) = \frac{1}{|\mathcal{W}|} \sum_{t \in \mathcal{W}} \frac{1}{|\Omega_{\text{flow}}|} \sum_{\mathbf{x} \in \Omega_{\text{flow}}} \|\tilde{\mathbf{u}}_t(\mathbf{x})\|_2 \quad (\uparrow), \quad (\text{S4})$$

$$\text{FP} = \min \left(\frac{\text{MCFF}(\mathcal{L})}{\text{MCFF}(\mathcal{E}) + \epsilon}, c \right) \quad (\uparrow), \quad (\text{S5})$$

clipped at a fixed c so that a vanishing early window cannot produce an unbounded ratio.

Drift leakage and attenuation. With F_{static} the mean flow magnitude over Ω_{static} and $F_{\text{raw}}, F_{\text{comp}}$ the mean raw and compensated magnitudes over Ω_{flow} , all in the late window,

$$\text{DLR} = \frac{F_{\text{static}}}{F_{\text{raw}} + \epsilon} \quad (\downarrow), \quad \text{DAR} = 1 - \frac{F_{\text{comp}}}{F_{\text{raw}} + \epsilon} \quad (\downarrow). \quad (\text{S6})$$

DLR is not a fraction and is deliberately unbounded: $\text{DLR} > 1$ indicates static support moving more than the region intended to move. DAR is stored signed and reported clipped to $[0, 1]$.

S2. Region Partition

Derivation. For every generated sequence we average optical-flow magnitude [8] over the leading 12% of sampled frame pairs and threshold it by Otsu’s method [4] into a dynamic region and its complement. The static region is then reduced by a fixed morphological erosion and a 4% frame border discarded; together these form the ignored transition band, so boundary pixels—where the partition is least certain and flow least reliable—contribute to neither region. The same procedure applies to both tracks.

Why the estimate is restricted to an early window. Deriving a partition from a sequence’s entire motion field would be circular, since accumulated background drift would itself be read as evidence that the affected region is dynamic. Restricting the estimate to the early window reduces circularity from late accumulated drift: at the 60s horizon the partition is fixed from the first 7.1s and is not defined by the late-window failure being measured. It nevertheless remains dependent on early checkpoint behavior and on the same flow estimator used downstream.

Factor	target	ρ	ρ_{rot}	off-target	admitted
fBD	drift	+1.00	+0.70	14%	✓
NBF	drift	+1.00	+0.60	6%	✓
MCFF-L	freeze	+1.00	n/a	14%	✓
FP	freeze	+0.80	n/a	21%	✓
DLR	translational leakage	+1.00	-0.80	5%	✓
DAR	drift	+0.90	+1.00	18%	✓
VBench DD	drift	+0.70	+0.70	2%	—

Table S1. Mechanistic validation and factor admission. ρ is signed target-family Spearman correlation; ρ_{rot} reports rotational response; off-target is the largest relative excursion under photometric and mask perturbations. Admission requires $\rho \geq 0.7$ and off-target response below 25%. Rotation is reported but not gated: DLR is inverted there, while fBD and DAR carry that case. VBench DD is a contrast row, not a candidate factor. Twelve clips per severity.

Limitations. The partition is obtained automatically rather than by annotation, so no inter-annotator agreement is reported, and for the image-conditioned track it is derived per generated sequence rather than once from the shared source image. The early window is a weaker guarantee than a first-frame annotation: a system that drifts within the first 12% of its rollout can influence its own partition. We therefore treat partitions as measured quantities and report the consequence directly—every measurement is recomputed under systematic erosion and dilation of the boundary, with the response in Table S1.

S3. Validation Protocol

Geometric severity is specified in a common physical unit—the mean pixel displacement a corruption induces over the frame by the end of the rollout—and each family’s parameter is derived from it using the frame geometry: a translation of d pixels, a rotation of $d/\bar{r}$ radians, or a zoom of $1 + d/\bar{r}$, where $\bar{r}$ is the mean distance from the frame center. This is what makes the geometric families comparable. A parameter in degrees is not commensurate with one in pixels: at the evaluation resolution one degree of rotation induces roughly 2.7px of mean displacement, so families compared by their native parameters differ in injected severity by more than an order of magnitude, and a factor can appear non-monotone simply because the corruption was too weak to clear the clip’s own baseline drift.

S4. Partition and Composition Robustness

Two questions decide whether the region partition can be trusted as a measurement basis: whether a system can be advantaged by receiving an easier mask, and whether the ordering depends on the one scene category the partition cannot represent.

The static region occupies a similar share of the frame for every audited system, between 0.61 and 0.66 on average, so no system is scored over a markedly larger or smaller support than another. Across systems, that area correlates negatively with feature-based drift. Over seven systems the correlation is not determined, but its sign is the one circularity would produce—a system that drifts within the estimation window enlarges its own dynamic region and shrinks the support its drift is measured over—and we report it as an open validity question rather than dismissing it on significance. The decisive test, recomputing every score under a region partition not derived from the same sequence, is stated as required future work in Sec. S2; we do not claim to have run it.

Removing precipitation, where rain and snow cross static support and a two-way partition must assign those pixels to one region by flow magnitude alone, leaves the normalized-background-flow ordering unchanged and moves only mid-field positions under feature-based drift.

S5. Paired Differences

Separation claims in the main text are made on the bootstrap distribution of per-prompt differences, not on whether two marginal intervals overlap. Every system sees the same prompts, so the paired difference is the statistic the design supports: overlapping marginal intervals can conceal a reliable paired difference, and non-overlapping ones can suggest a difference that is not there.

S6. Generation Configuration

Every text-conditioned checkpoint was loaded by the same inference program and configuration. Fixed before the sweep to give the distilled checkpoint family a shared four-step budget, it specifies a four-step denoising schedule with indices 1000/750/500/250 warped by the released scheduler at timestep shift 5.0, six frames per block, and seed 0. All T2V outputs are 832×480 at 16 fps. Within a horizon, only checkpoint weights and their recorded regular/EMA selection vary. The YAML retains a `guidance_scale`: 5.0 field, but the selected few-step CausalInferencePipeline is conditional-only and never reads it; classifier-free guidance is therefore not an applied parameter in this sweep.

We state plainly what this does and does not license. It is a recorded common-configuration comparison, reproducible because the inference path is uniform and recorded. It is not an audit of each method under its authors’ released defaults, and we no longer describe it as one. The cost is real: several of these checkpoints

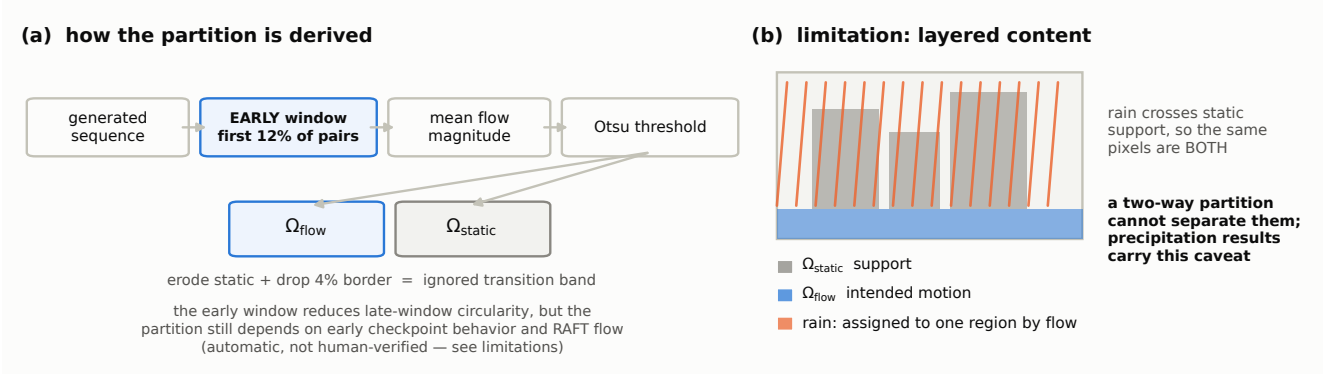


Figure S1. Region-partition protocol. (a) The partition is derived automatically from early-window flow, with an ignored transition band at the boundary. (b) A two-way partition cannot represent layered content such as rain crossing static support, which is the scope limit stated in the main paper.

System	static-region area
CausVid	0.663 ± 0.073
Reward-Forcing	0.661 ± 0.060
Self-Forcing	0.651 ± 0.087
Infinite-Forcing	0.650 ± 0.090
LongLive	0.644 ± 0.095
Rolling-Forcing	0.616 ± 0.107
Causal-Forcing	0.614 ± 0.086
Does mask area predict the drift score?	
Spearman(area, fBD) over 7 systems	-0.75, exact $p = 0.066$
Ordering with precipitation excluded (3 of 23 prompts)	
fBD ordering	best and worst unchanged
NBF ordering	identical

Table S2. Partition and composition robustness. The static region occupies a similar share of the frame for every audited system, so no system is scored over a much larger or smaller support than another. The area does correlate negatively with drift across systems; over seven systems that correlation is not determined, but its sign is the one circularity would produce—a system that drifts early enlarges its own dynamic region—and we flag it as an open validity question rather than dismissing it. Removing precipitation, the category a two-way partition cannot represent, leaves the normalized-background-flow ordering unchanged.

are distilled for a specific step budget, and evaluating them at a common four-step schedule does not observe them at their own operating point. Results are therefore statements about these checkpoints under this configuration. Re-running the sweep at each method’s released defaults, and reporting how much the ordering moves, is the natural next version of this audit.

The image-conditioned roster contains two distinct settings. Unmarked rows are recorded outputs produced by model-specific released pipelines. Rows marked [†] use the common long-horizon rollout wrapper with six frames per block and a fixed seed; unlike T2V, denoising schedules are checkpoint-specific. These wrapper parameters do not apply to unmarked rows. The marked rows are read as deployment sen-

sitivity and are never ranked against the recorded released-pipeline outputs.

S7. Measurement Coverage

Coverage is counted from valid per-video measurements rather than from the presence of a results file, so a run that terminated early is reported as partial. Separately, fBD is feature-based [6] and declines to report where too few repeatable keypoints survive; those abstentions are tabulated rather than folded silently into a smaller n .

Comparison	n	mean diff.	95% CI
Infinite-Forcing – Causal-Forcing	23	-83.98	[-102.59, -65.52]
Infinite-Forcing – LongLive	23	-10.22	[-14.72, -6.18]
Infinite-Forcing – Self-Forcing	23	-9.07	[-13.29, -5.53]
Infinite-Forcing – Rolling-Forcing	23	-7.58	[-10.38, -5.12]
Infinite-Forcing – CausVid	23	-5.62	[-9.61, -2.10]
Infinite-Forcing – Reward-Forcing	23	-4.86	[-7.48, -2.69]

Table S3. Paired differences in NBF, 60s horizon. Every system is evaluated on the same prompts, so separation is judged on the bootstrap distribution of per-prompt differences (10k resamples, fixed seed) rather than on whether two marginal intervals happen to overlap—overlapping intervals can hide a reliable paired difference, and non-overlapping ones can suggest a difference that is not. 6 of 6 intervals exclude zero.

System	setting	resolution	fps	horizons
CausVid	common	832×480	16	5s, 60s, 120s, 240s
Self-Forcing	common	832×480	16	5s, 60s, 120s, 240s
Infinite-Forcing	common	832×480	16	5s, 60s, 120s, 240s
Rolling-Forcing	common	832×480	16	5s, 60s, 120s, 240s
Reward-Forcing	common	832×480	16	5s, 60s, 120s, 240s
LongLive	common	832×480	16	5s, 60s, 120s, 240s
Causal-Forcing	common	832×480	16	5s, 60s, 120s, 240s

Table S4. Recorded generation configuration (T2V). Every audited system was run from its released checkpoint. Output formats are normalized at measurement time: NBF is expressed per second and per frame width, fBD as a percentage of the frame diagonal, and every sequence is resampled to a common flow-estimation rate. All rows use the recorded common T2V configuration: a four-step denoising schedule with released-scheduler warping, six frames per block, and seed 0. These are checkpoint results under this common configuration, not reconstructions of the methods’ released inference procedures.

178 S8. Complete Audit, Both Tracks

179 The main paper reports the text-conditioned track at
 180 60s, the tier whose sample supports method-level separation.
 181 Both tracks appear here at every horizon, with
 182 uncertainty from a percentile bootstrap over items [1].
 183 For the image-conditioned track, dagger-marked wrapper
 184 rows use a common long-horizon rollout wrapper
 185 rather than a released pipeline, so their scores are read
 186 as deployment sensitivity rather than as the published
 187 method’s performance.

188 S9. Aggregation Sensitivity

189 The evaluation set is dominated by one flow medium,
 190 so the audit must be shown not to be a report on that
 191 medium alone. The main paper aggregates by averaging
 192 over prompts. The alternative, weighting each
 193 scene category equally, is not preferable here: at the
 194 principal horizon three of the five populated categories
 195 contain two prompts each, so a macro-average would
 196 give a two-prompt cell the same influence as a fourteen-
 197 prompt one and inflate variance in exchange for nominal
 198 balance. We therefore report both and state what
 199 changes.

200 The best- and worst-ranked systems are the same

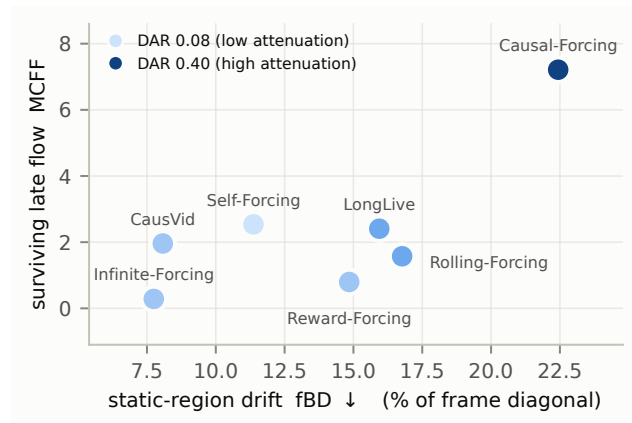


Figure S2. Operating regimes under the recorded common T2V configuration. Each checkpoint in the static-drift–surviving-flow plane. Color encodes low-to-high attenuation of measured dynamic-region flow after global-motion compensation, not a share attributable to drift: the quantity is signed, and values are shown clipped to [0, 1]. The view exposes trade-offs; it is not a leaderboard.

under either aggregation, so the contrast the paper
 draws does not depend on the choice. Intermediate
 ranks do move, which is why no ordinal claim is made

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System	setting	resolution	fps	horizons
Causal-Forcing++ (2-step, frame-wise)	released	832×480	16	5s, 60s, 120s, 240s
Wan2.1-I2V-14B-480P	released	832×464	16	5s
Wan2.2-I2V-A14B	released	832×464	16	5s
LTX-Video 13B-0.9.8-distilled	released	832×480	24	5s
Causal-Forcing++ (1-step, frame-wise)	released	832×480	16	60s, 120s
Causal-Forcing++ (2-step) [†]	wrapper	832×480	16	5s, 60s, 120s, 240s
Causal-Forcing++ (1-step) [†]	wrapper	832×480	16	5s, 60s, 120s, 240s
Causal-Forcing (frame-wise) [†]	wrapper	832×480	16	5s, 60s, 120s, 240s
Self-Forcing [†]	wrapper	832×480	16	5s, 60s, 120s, 240s
CausVid [†]	wrapper	832×480	16	5s, 60s, 120s, 240s

Table S5. Recorded generation configuration (I2V). Every audited system was run from its released checkpoint. Output formats are normalized at measurement time: NBF is expressed per second and per frame width, fBD as a percentage of the frame diagonal, and every sequence is resampled to a common flow-estimation rate. Unmarked rows are recorded outputs from model-specific released pipelines and carry the setting label released. [†] rows use the common long-horizon rollout wrapper (six frames per block, fixed seed; checkpoint-specific denoising schedules); those parameters do not apply to unmarked rows, and the two settings are not ranked together.

track	horizon	measured	fBD abstained	rate
I2V	120s	140	0	0.0%
I2V	240s	30	0	0.0%
I2V	5s	90	0	0.0%
I2V	60s	210	2	1.0%
T2V	120s	42	0	0.0%
T2V	240s	28	0	0.0%
T2V	5s	48	0	0.0%
T2V	60s	162	0	0.0%

Table S6. Public-roster measurement coverage and fBD abstention. fBD is feature-based and declines to report when too few repeatable keypoints survive in the static region. Public-roster abstentions are confined to a desert dust-storm scene whose flat, dust-obscured ground offers little stable structure to track. The remaining factors are reported for those clips as usual; only fBD is withheld.

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205 S10. Whole-Frame and SNF-Bench Rank Cor-
206 relation

207 S11. Deployment Sensitivity

208 S12. Signed Drift Attenuation

209 Drift attenuation is stored signed and reported clipped
210 to $[0, 1]$. Negative values are informative rather than
211 anomalous: where local flow opposes the estimated
212 global field, compensation increases the measured mag-
213 nitude. We report their incidence rather than suppress-
214 ing it.

S13. Evaluation-Set Composition 215

S14. Exploratory Perceptual Pilot 216

This section supports no perceptual-validity claim. It reports an exploratory pilot that did not establish agreement between human judgments and the factor orderings, and it is included because the reason it failed is informative and because the protocol is what a powered study should start from. Nothing here ranks systems, and no result from it enters any factor, table or figure elsewhere in this work. 217
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The pilot ran 24 pairs per rater—twelve on each axis—with eight raters, of whom four were independent of the authors and four were the authors’ own sessions, reported separately. After majority voting across raters, three pairs on the static axis and five on the persistence axis remained decided. Those counts, not the percentages they produce, are the honest summary of this run. 223
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The question the pilot asked is whether independent judges order clip pairs the way the factors do. That is metric validation rather than preference, so a small sample would have been survivable had the pairs been constructed correctly; as Sec. S14 reports below, they were not. 233
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Ethics. Raters were volunteers drawn from the authors’ institution and immediate colleagues, and four of the eight sessions were the authors’ own pilot runs, reported separately and excluded from the primary analysis. Participation was voluntary and uncompensated, no personal data beyond a self-chosen identifier was collected, and no clip depicts an identifiable person. The study involved no intervention and no sensitive 239
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Method	status	setting	5s	60s	120s	240s
CausVid	public	common	V6 T6 B	V23 T23 B	V6 T6 B	V4 T4 B
Self-Forcing	public	common	V6 T6 B	V23 T23 B	V6 T6 B	V4 T4 B
Infinite-Forcing	public	common	V6 T6 B	V23 T23 B	V6 T6 B	V4 T4 B
Rolling-Forcing	public	common	V6 T6 B	V24 T24 B	V6 T6 B	V4 T4 B
Reward-Forcing	public	common	V12 T12 B	V23 T23 B	V6 T6 B	V4 T4 B
LongLive	public	common	V6 T6 B	V23 T23 B	V6 T6 B	V4 T4 B
Causal-Forcing	public	common	V6 T6 B	V23 T23 B	V6 T6 B	V4 T4 B

Table S7. Asset and score coverage for the audited public systems. V_n is the number of available video assets, T_n the number of valid SNF task-metric records, and B denotes available VBench scores. T_n/N marks entries where only n of N videos hold usable measurements. V counts generated assets, whereas audit n counts unique frozen-manifest prompt items after duplicate prompt outputs are collapsed. Reward-Forcing has 12 assets over six 5s prompts; Rolling-Forcing has 24 assets over 23 60s prompts.

Method	fBD↓	NBF↓	MCFF-E	MCFF-L↑	FP↑	DLR↓	DAR↓
5s horizon ($n = 6$)							
CausVid	2.08	6.56	1.44	2.91	1.24	0.26	0.07
Self-Forcing	0.68	10.8	7.70	4.35	0.76	0.49	0.03
Infinite-Forcing	0.24	2.32	1.23	0.89	1.08	0.21	0.09
Rolling-Forcing	3.51	12.1	3.10	3.69	1.32	0.48	0.17
Reward-Forcing	1.10	5.21	0.63	0.58	1.37	0.44	0.28
LongLive	1.91	7.31	1.25	2.27	1.16	0.59	0.32
Causal-Forcing	9.55	31.0	5.32	3.01	1.13	0.62	0.37
60s horizon ($n = 23$)							
CausVid	8.08	9.27	2.87	1.96	0.67	0.45	0.14
Self-Forcing	11.4	12.7	10.4	2.54	0.44	0.50	0.14
Infinite-Forcing	7.75	3.65	1.17	0.29	0.63	0.55	0.14
Rolling-Forcing	16.8	11.2	3.18	1.57	0.75	0.54	0.18
Reward-Forcing	14.8	8.51	3.81	0.80	0.49	0.60	0.17
LongLive	15.9	13.9	6.46	2.41	0.75	0.59	0.17
Causal-Forcing	22.4	87.6	9.84	7.21	0.83	0.94	0.43
120s horizon ($n = 6$)							
CausVid	6.15	3.42	0.36	0.52	1.06	0.41	0.13
Self-Forcing	21.2	6.80	4.85	0.82	0.85	0.75	0.15
Infinite-Forcing	16.1	2.60	0.50	0.32	0.64	0.57	0.21
Rolling-Forcing	23.8	10.6	3.33	0.77	0.38	0.82	0.22
Reward-Forcing	20.0	11.6	2.05	0.69	0.39	0.70	0.12
LongLive	22.5	9.66	5.10	1.19	0.81	0.84	0.23
Causal-Forcing	21.8	130	10.5	10.7	0.90	0.81	0.50
240s horizon ($n = 4$)							
CausVid	7.23	13.4	0.42	0.87	0.99	0.85	0.08
Self-Forcing	20.0	17.4	3.09	0.50	0.15	2.26	0.20
Infinite-Forcing	20.0	4.15	0.33	0.23	1.02	1.08	0.15
Rolling-Forcing	23.7	15.4	2.15	0.97	0.46	0.78	0.18
Reward-Forcing	26.4	11.4	1.09	3.55	1.53	0.50	0.07
LongLive	23.1	11.8	1.52	1.32	1.20	1.38	0.31
Causal-Forcing	18.9	102	4.72	6.90	1.21	0.73	0.58

Table S8. Complete T2V audit, all four horizons. The main paper reports the 60s and 120s panels; the 5s tier is an initialization check and 240s a diagnostic extreme. Every row is computed under the same frozen specification, so panels are comparable within a horizon; prompt sets differ across horizons, so columns are not comparable between panels. Header n is the maximum valid item count within each horizon; factor-specific abstentions are in Table S6. Units: fBD is % frame diagonal; NBF is 10^{-3} frame widths/s; MCFF is px/sampled interval; FP is unitless. FP is the mean of per-video clipped late/early ratios and therefore need not equal the ratio of the displayed aggregate MCFF-L and MCFF-E means.

Method	setting	fBD↓	NBF↓	MCFF-E	MCFF-L↑	FP↑	DLR↓	DAR↓
5s horizon ($n = 10$)								
Causal-Forcing++ (2-step)	wrapper	3.69	10.0	1.06	0.61	0.78	0.67	0.32
Causal-Forcing++ (1-step)	wrapper	5.28	11.4	1.05	0.47	0.64	0.66	0.29
Causal-Forcing (frame-wise)	wrapper	16.1	45.3	8.48	2.79	0.43	0.68	0.29
Causal-Forcing++ (2-step, frame-wise)	released	1.22	10.6	2.23	0.78	0.47	0.68	0.20
Self-Forcing	wrapper	9.66	5.74	0.69	0.26	0.70	0.74	0.21
CausVid	wrapper	15.2	10.8	0.76	0.87	0.98	1.04	0.10
Wan2.1-I2V-14B-480P	released	5.01	21.3	6.67	5.58	1.09	0.52	0.18
Wan2.2-I2V-A14B	released	6.31	25.3	4.33	9.09	0.95	0.50	0.27
LTX-Video 13B-0.9.8-distilled	released	6.65	17.8	3.59	3.48	0.92	0.39	0.24
60s horizon ($n = 30$)								
Causal-Forcing++ (2-step)	wrapper	9.18	5.97	0.85	0.60	0.86	0.67	0.25
Causal-Forcing++ (1-step)	wrapper	5.34	7.05	0.86	0.57	0.77	0.64	0.28
Causal-Forcing (frame-wise)	wrapper	21.1	36.4	2.80	2.69	0.89	0.80	0.35
Causal-Forcing++ (2-step, frame-wise)	released	9.94	6.91	1.30	0.78	1.02	0.75	0.17
Self-Forcing	wrapper	17.3	5.52	1.50	0.69	0.89	0.67	0.18
CausVid	wrapper	19.9	3.31	0.58	0.41	0.81	0.57	0.14
Causal-Forcing++ (1-step, frame-wise)	released	10.4	5.25	0.53	0.52	1.01	0.83	0.22
120s horizon ($n = 20$)								
Causal-Forcing++ (2-step)	wrapper	9.65	8.81	0.89	0.82	0.93	0.58	0.28
Causal-Forcing++ (1-step)	wrapper	7.26	9.39	0.84	0.66	0.86	0.62	0.31
Causal-Forcing (frame-wise)	wrapper	19.9	52.7	4.82	5.06	1.08	0.67	0.24
Causal-Forcing++ (2-step, frame-wise)	released	11.4	17.3	1.89	1.54	1.01	0.75	0.09
Self-Forcing	wrapper	23.0	10.3	2.41	0.76	0.45	0.66	0.26
CausVid	wrapper	19.7	6.04	1.15	0.83	0.67	0.48	0.14
Causal-Forcing++ (1-step, frame-wise)	released	13.6	7.39	1.01	0.35	0.64	0.77	0.33
240s horizon ($n = 5$)								
Causal-Forcing++ (2-step)	wrapper	10.4	4.83	0.53	0.59	1.05	0.55	0.15
Causal-Forcing++ (1-step)	wrapper	6.42	6.66	0.56	0.51	0.83	0.67	0.30
Causal-Forcing (frame-wise)	wrapper	19.8	31.5	3.08	2.28	0.74	0.69	0.45
Causal-Forcing++ (2-step, frame-wise)	released	6.32	3.95	1.23	0.68	0.67	0.83	0.02
Self-Forcing	wrapper	22.3	5.74	1.48	1.59	0.81	0.71	0.08
CausVid	wrapper	18.4	2.08	0.81	0.79	0.86	0.44	0.11

Table S9. Complete I2V audit, all four horizons. The main paper reports the 60s deployment panel; the 5s tier is an initialization check and 240s a diagnostic extreme. Every row is computed under the same frozen specification, so panels are comparable within a horizon; prompt sets differ across horizons, so columns are not comparable between panels. Header n is the maximum valid item count within each horizon; factor-specific abstentions are in Table S6. Units: fBD is % frame diagonal; NBF is 10^{-3} frame widths/s; MCFF is px/sampled interval; FP is unitless. FP is the mean of per-video clipped late/early ratios and therefore need not equal the ratio of the displayed aggregate MCFF-L and MCFF-E means.

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material. Because a powered study will recruit outside this group, we will obtain the institutional review or documented exemption that such recruitment requires before running it.

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Questions. One forced choice per axis, with an explicit third option: static stability—in which clip does the background stay more fixed in place—and motion persistence—in which clip does the intended motion keep going rather than fade or freeze. Judges are never asked which output is better, or which looks more natural. “Better” conflates the two failures the benchmark

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separates, and naturalness is outside the stated scope, so either question would produce a result the paper could not own.

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Pairs. A pair is two systems’ outputs on the same prompt at the same horizon, never across prompts. Pairs are stratified by the relative gap in the axis’s own factor: clear-gap pairs, where the factor separates the two outputs, carry the agreement claim, while near-tie pairs test the opposite property—that where the factors do not separate, judges also cannot tell the clips apart. Presentation side is randomized per trial and re-

System	prompt mean	rank	macro mean	rank
NBF				
Infinite-Forcing	3.65	1	4.12	1
Reward-Forcing	8.51	2	10.11	4 [†]
CausVid	9.27	3	9.93	3
Rolling-Forcing	11.23	4	11.32	5 [†]
Self-Forcing	12.72	5	8.98	2 [†]
LongLive	13.86	6	14.69	6
Causal-Forcing	87.62	7	61.70	7
fBD				
Infinite-Forcing	7.75	1	6.10	1
CausVid	8.08	2	9.51	3 [†]
Self-Forcing	11.37	3	9.27	2 [†]
Reward-Forcing	14.85	4	15.61	5 [†]
LongLive	15.94	5	14.93	4 [†]
Rolling-Forcing	16.77	6	18.85	6
Causal-Forcing	22.44	7	23.49	7

Table S10. Sensitivity of the audit to category aggregation. Prompt means, as reported in the main paper, against category macro-averages, for the two static-fidelity factors at the principal horizon. [†] marks a system whose rank changes. The best- and worst-ranked systems are identical under both, so the paper’s headline contrast does not depend on the choice; intermediate positions do change, which is why no ordinal claim is made about the middle of the field.

generic metric	fBD	NBF	MCFF	DLR
VB-DD	+0.75	+0.93	+0.89	+0.36
VB-bg	-0.64	-0.71	-0.75	-0.50
VB-smooth / VB-flick	-0.96	-0.82	-0.61	-0.64

Table S11. Rank disagreement at 60s. Spearman correlation over $n = 7$ method means—descriptive, not inferential. Positive correlation between apparent motion and background flow is the effect the benchmark isolates. Smoothness and flickering rank these systems identically and share a row.

solved back to a system identity only after the response is recorded; system names are never shown.

Statistics. The unit of analysis is the pair, majority-voted across raters, and intervals are percentile bootstrap resampling pairs. Responses are clustered inside rater and inside pair, so an interval computed over responses would claim precision the design does not have. We report inter-rater agreement so the reader can see whether the raters are signal or noise, and we run no significance test the sample cannot power: the claim is descriptive agreement, and saying so plainly is preferable to a p -value that would not survive scrutiny.

Outcome of the human arm: the protocol executed end to end; agreement is not yet estimable. Eight raters each completed all 24 trials, four of them independent of the authors; the remaining four were pilot sessions

and are reported separately rather than pooled. Engagement was genuine—median deliberation ran from 23 to 233 seconds per trial—and the “can’t tell” option was used deliberately, on 30% of responses, rather than avoided.

The agreement estimate, however, does not survive its own error bars. Majority voting across raters leaves three decided pairs on the static axis and five on the persistence axis, and the corresponding intervals span most of the unit range. Inter-rater agreement is low ($\alpha = 0.18$ and 0.24 on the two axes), and the direction of the estimate reverses depending on which rater subset is used. We therefore report the counts and the interval rather than a headline percentage, and we do not claim that perceptual agreement has been established.

The cause is a property of the pair set rather than of the raters, and it is measurable. The pairs were stratified by the gap in each axis’s own factor but were not isolated against the other axis. On the static question the median relative gap in the target factor is 0.29 while the median gap in the persistence factor is 0.58 : the clips shown differ about twice as much on the axis the rater was told to ignore as on the one being asked about, and on half the pairs both axes differ substantially. A rater seeing one clip drift and the other stall cannot report which axis drove the answer, which is exactly the condition that produces low inter-rater agreement and near-chance ordering. A powered study therefore requires axis-isolated pairs—large gap in the

Method	Generic evidence	SNF-Bench evidence	Interpretation
CausVid	VB-DD rank 6/7	NBF rank 3/7, MCFF-L rank 4/7	stable sequence with decaying intended motion
Causal-Forcing	VB-DD rank 1/7	NBF rank 7/7, MCFF-L rank 1/7	high apparent motion accompanied by high static-region motion under the recorded common configuration
Infinite-Forcing	VB-DD rank 7/7	NBF rank 1/7, MCFF-L rank 7/7	stable sequence with decaying intended motion
LongLive	VB-DD rank 3/7	NBF rank 6/7, MCFF-L rank 3/7	high apparent motion accompanied by high static-region motion under the recorded common configuration
Reward-Forcing	VB-DD rank 5/7	NBF rank 2/7, MCFF-L rank 6/7	stable sequence with decaying intended motion
Self-Forcing	VB-DD rank 2/7	NBF rank 5/7, MCFF-L rank 2/7	high apparent motion accompanied by high static-region motion under the recorded common configuration

Table S12. Interpretation changes between generic whole-frame metrics and SNF-Bench at 60 s. Neither evaluation is labeled correct; the table identifies information hidden by whole-frame aggregation. Cases are selected by a fixed rank-gap rule applied to all public methods. Rolling-Forcing does not satisfy the fixed three-rank-gap inclusion criterion and is therefore absent.

Checkpoint	horizon	Δ fBD	Δ NBF	Δ MCFF-L	Δ DLR
Causal-Forcing++ (2-step)	5s	+2.47	-0.61	-0.17	-0.01
Causal-Forcing++ (2-step)	60s	-0.76	-0.95	-0.17	-0.08
Causal-Forcing++ (2-step)	120s	-1.71	-8.46	-0.73	-0.17
Causal-Forcing++ (2-step)	240s	+4.06	+0.88	-0.09	-0.28
Causal-Forcing++ (1-step)	60s	-5.05	+1.80	+0.05	-0.20
Causal-Forcing++ (1-step)	120s	-6.34	+2.00	+0.31	-0.14

Table S13. Deployment sensitivity. Change from each checkpoint’s own released setting to the common long-horizon wrapper, for the two checkpoints evaluated both ways. Absolute wrapper scores are never read as the published method’s performance.

315	target factor, small gap in the other—and that constraint, not rater count, is the first thing to fix.	We report none.	333
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317	Outcome of the vision–language arm: negative, with a located cause. The same pairs and the same wording were put to an open-weight judge [5] at temperature zero, each pair asked in both presentation orders. Two controls precede any reading of its answers. Shown one clip against itself, where the only correct answer is that they are the same, it answered correctly on six of six. Shown a frozen single frame against a real sequence—the largest difference available on the motion axis—its verdict followed the swapped clip on only two of six. Presentation bias is absent: over 48 queries the two slots were chosen 8 and 12 times. The judge is therefore well behaved and still cannot do the task: it answered “indistinguishable” on most pairs the factors separate clearly, leaving one and two decided pairs on the two axes, which is far too few to support an agreement rate.	The controls locate the cause in the judge rather than in the factors or the prompt. A model that cannot separate a still image from moving water is not positioned to separate two rollouts that differ subtly. This says that an open-weight judge at this scale cannot presently substitute for human raters on these axes. It does not say that stronger judges would fail, and we make no claim about them: contemporary generation benchmarks that report successful automated judging use frontier proprietary models [2, 7] or scorers fine-tuned for the task [3], and we have no evidence either way about those here.	334 335 336 337 338 339 340 341 342 343 344 345
320		Status. Both arms are reported as preliminary and neither supports an agreement claim. What the pilot does establish is that the protocol executes end to end, that raters engage with the axis questions rather than defaulting to a side, and that the binding constraint	346 347 348 349 350

Method	track	dur	n	negative	rate	most negative
CausVid	i2v	120s	20	1	5.0%	-0.055
CausVid	i2v	5s	10	5	50.0%	-1.076
CausVid	i2v	60s	30	2	6.7%	-0.084
Causal-Forcing++ (1-step)	i2v	120s	20	1	5.0%	-0.027
Causal-Forcing (frame-wise)	i2v	120s	20	3	15.0%	-0.171
Causal-Forcing++ (2-step)	i2v	5s	10	1	10.0%	-0.164
Causal-Forcing++ (1-step, frame-wise)	i2v	60s	30	3	10.0%	-1.420
Causal-Forcing++ (2-step, frame-wise)	i2v	120s	20	5	25.0%	-0.241
Causal-Forcing++ (2-step, frame-wise)	i2v	240s	5	1	20.0%	-0.651
Causal-Forcing++ (2-step, frame-wise)	i2v	5s	10	1	10.0%	-0.288
Causal-Forcing++ (2-step, frame-wise)	i2v	60s	30	1	3.3%	-0.001
LTX-Video 13B-0.9.8-distilled	i2v	5s	10	2	20.0%	-0.018
Self-Forcing	i2v	120s	20	2	10.0%	-0.046
Self-Forcing	i2v	240s	5	1	20.0%	-0.136
Self-Forcing	i2v	60s	30	4	13.3%	-0.578
Wan2.1-I2V-14B-480P	i2v	5s	10	1	10.0%	-0.255
Wan2.2-I2V-A14B	i2v	5s	10	1	10.0%	-0.335
CausVid	t2v	240s	4	1	25.0%	-0.384
CausVid	t2v	5s	6	2	33.3%	-0.093
Causal-Forcing	t2v	5s	6	2	33.3%	-0.273
Causal-Forcing	t2v	60s	23	3	13.0%	-0.640
Infinite-Forcing	t2v	240s	4	1	25.0%	-0.158
Infinite-Forcing	t2v	60s	23	4	17.4%	-0.359
LongLive	t2v	120s	6	1	16.7%	-1.076
LongLive	t2v	5s	6	1	16.7%	-0.139
LongLive	t2v	60s	23	2	8.7%	-0.051
Reward-Forcing	t2v	120s	6	1	16.7%	-0.116
Reward-Forcing	t2v	5s	12	2	16.7%	-0.145
Reward-Forcing	t2v	60s	23	3	13.0%	-0.317
Rolling-Forcing	t2v	120s	6	3	50.0%	-0.353
Rolling-Forcing	t2v	5s	6	1	16.7%	-0.068
Rolling-Forcing	t2v	60s	24	2	8.3%	-0.038
Self-Forcing	t2v	120s	6	2	33.3%	-0.130
Self-Forcing	t2v	240s	4	1	25.0%	-0.153
Self-Forcing	t2v	5s	6	1	16.7%	-0.046
Self-Forcing	t2v	60s	23	4	17.4%	-1.117

Table S14. Incidence of negative DAR, i.e. clips where global-motion compensation increased measured dynamic-region flow. 21 further entries had no negative values and are omitted.

track/duration	n	river_stream	ocean_waves	precipitation	fire_smoke	lava_volcanic	windborne	cats
i2v/5s	10	2	.	3	2	1	2	5
i2v/60s	30	9	4	9	1	3	4	6
i2v/120s	20	6	3	3	3	1	4	6
i2v/240s	5	1	2	1	.	1	.	4
t2v/5s	6	3	1	1	1	.	.	4
t2v/60s	23	14	2	3	2	.	2	5
t2v/120s	6	4	.	1	.	.	1	3
t2v/240s	4	2	.	1	.	.	1	3

Table S15. Scene-category balance of the evaluation set, by track and horizon. A dot marks a category absent from that cell.

on a powered study is pair construction rather than rater count: axis-isolated pairs, a near-tie stratum, and raters recruited outside the authors. Protocol, pair manifest and judge prompt are released verbatim so the check can be repeated and enlarged. To restate the scope without qualification: what is released here is a documented, executable protocol and a diagnosis of why this run of it failed. No conclusion about perceptual agreement, in either direction, should be drawn from the counts in this section. They are reported so the pilot is auditable, not because they support an inference.

S15. Additional Qualitative Material

Figure 1 selection rule. For each prompt represented by at least six systems, the support-preserving candidate minimizes $\text{fBD} - 2\text{MCFF-L}$, the drift candidate maximizes fBD , and the stagnation candidate mini-

mizes MCFF-L . Among prompts with three distinct candidates, the selected prompt maximizes $(\text{fBD}_{\text{drift}} - \text{fBD}_{\text{support}})/(1 + 40|\text{DD}_{\text{drift}}^* - \text{DD}_{\text{support}}^*|)$.

Axis	Judge	agrees with factor	called indistinguishable	pairs	inter-rater α
Static stability (fBD)	human	– (3 decided)	–	8	0.18
	vision–language	– (1 decided)	1/1	7	–
Motion persistence (MCFF-L)	human	4/5 decided pairs	–	8	0.24
	vision–language	– (2 decided)	–	3	–

Table S16. Preliminary perceptual pilot for the factor axes. Each judge answers one question per axis and never which output is better. agree is agreement with the factor’s ordering on pairs the factor separates clearly, over pairs the judge decided; can’t tell is the share of near-tie pairs called indistinguishable, which is the correct answer there. The unit is the pair, majority-voted across raters. A dash marks a cell with too few decided pairs to carry a rate; the count is given instead. This is a preliminary check on a measurement protocol, not a powered study, and no system is ranked by it.

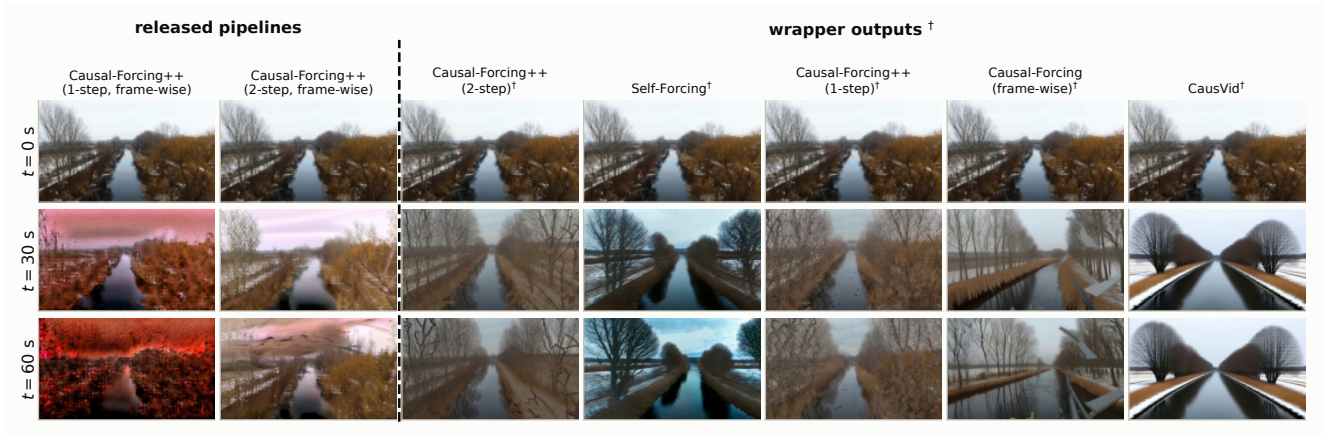


Figure S3. Image-conditioned systems on one shared source image. Every system begins from the identical frame, so divergence over the rollout is attributable to the system rather than to a differently imagined scene—which is not true of the text-conditioned track, where each system authors its own layout. Released-pipeline and wrapper outputs are visibly separated and ordered by increasing fBD only within their respective groups; [†] marks a wrapper output, and no cross-setting ordering is intended.

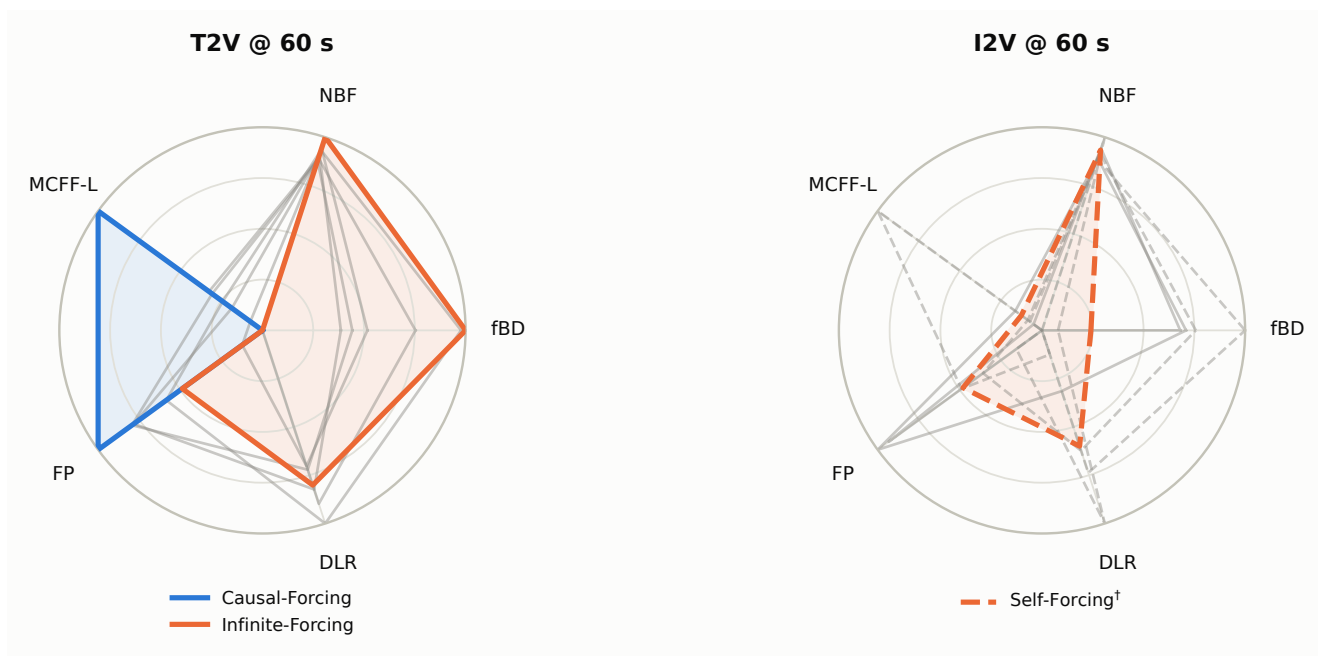


Figure S4. Factor profile per track. Each factor is rescaled across the audited systems and oriented so that outward is always better; plotting raw values would place “most drift” and “most surviving motion” both outward and the shape would carry no meaning. No system encloses the others on either track, which is the conclusion the operating-regime view reaches, shown as a profile. On the image-conditioned panel, dashed outlines marked [†] are rows run under the common long-horizon wrapper rather than a released pipeline: they are shown for shape, and are not peers of the solid released-pipeline outlines. The rescaling is relative to the audited roster, so a profile is descriptive of this set and would change if the set changed.



Figure S5. Stated scope boundaries. (a) A two-way partition cannot separate precipitation from the support it crosses. (b) A system can attain near-best static fidelity precisely because its intended motion has also stopped. (c) High leakage with low attenuation marks static-region motion for which the fitted global similarity model attenuates little measured flow; what the residual consists of is not identified by these measurements.



Figure S6. Every audited text-conditioned system on a second prompt. Rows show $t = 0/30/60$ s; columns are ordered by increasing static-region drift. This sample excludes the main-paper prompt and maximizes cross-system fBD spread over the remaining prompts.

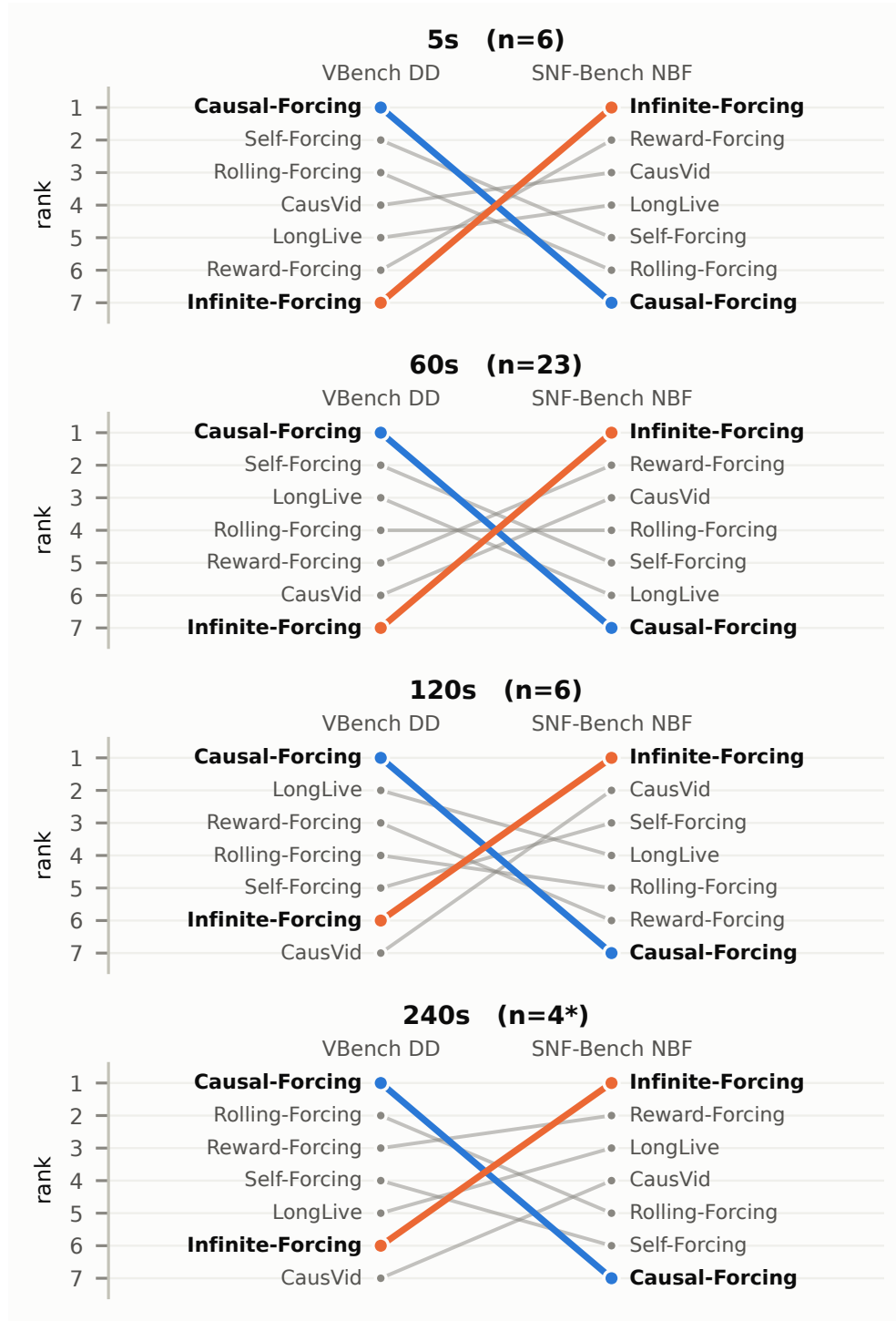


Figure S7. Rank disagreement at every audited horizon. Rank under a whole-frame motion score against rank under normalized background flow. Both quantities are independent of the global-motion estimator; prompt counts are given per panel. An asterisk marks a horizon where Dynamic Degree is available only as a method-level V-Bench aggregate, suitable for descriptive ranks but not for item-level uncertainty.

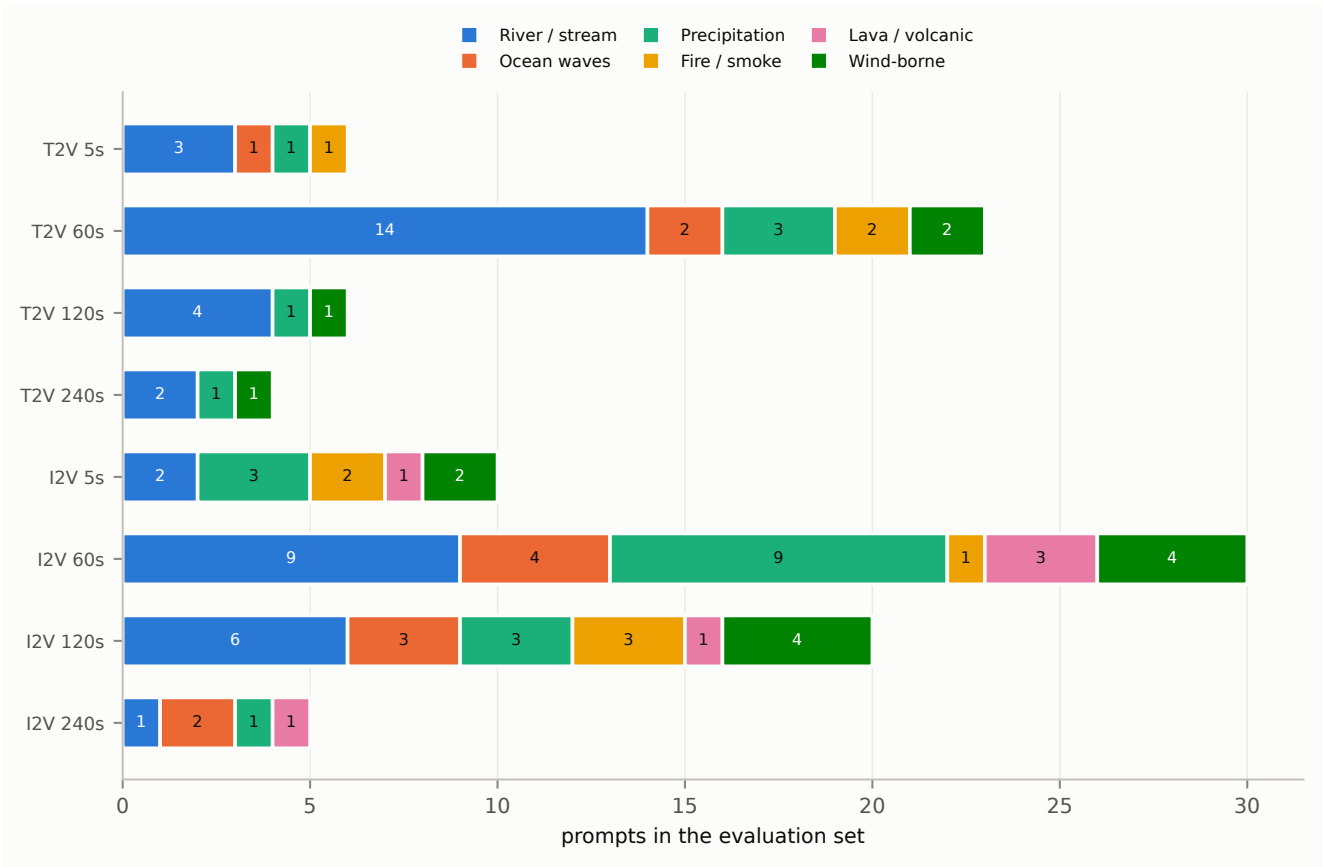


Figure S8. Composition of the evaluation set by scene category, in prompts per category at each track and horizon.

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